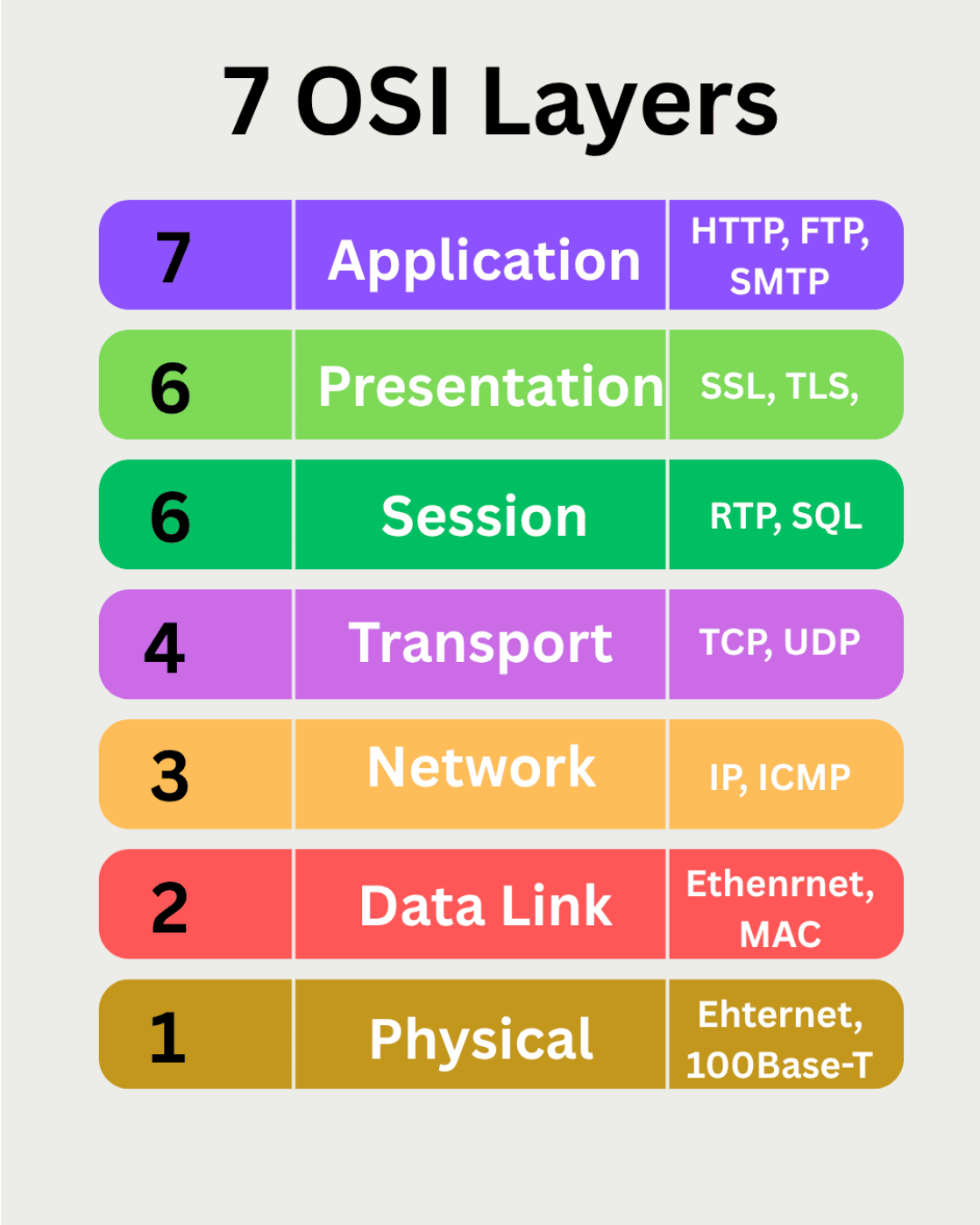
1. Draw a diagram of the OSI 7-layer model and label each layer with its correct name and a real-world example of a device or protocol that operates at that layer (e.g., Switch at Data Link, HTTP at Application).

   Ans:-



2. Match the following networking scenarios to the correct OSI layer: a) Encrypting data before sending, b) Routing packets across networks, c) Converting data to electrical signals, d) Opening a web page in your browser, e) Detecting and correcting errors in frames.

Hint: Write your answers as 'Scenario: Layer Name (Layer Number)'.

Ans:-

1. **Presentation Layer (6)** : Encrypting data before sending:
2. **Network Layer (3)** : Routing packets across networks:
3. **Physical Layer (1)** :Converting data to electrical signals
4. **Application Layer (7)** : Opening a web page in your browser
5. **e) Data Link Layer (2):** Detecting and correcting errors in frames:

3. Pick any app you use daily (like WhatsApp, Instagram, or Zomato) and describe how at least three different OSI layers are involved when you send or receive data in that app.

Ans:-

App Use: WhatsApp

- When I send a message to my friend on WhatsApp, different OSI layers work together to deliver the message.

* Application Layer (Layer 7): WhatsApp allows me to type and send messages, photos, and videos using the app.

* Transport Layer (Layer 4): TCP or UDP-based transport mechanisms help transfer data between devices and servers. WhatsApp uses different transport protocols depending on the type of communication.

* Network Layer (Layer 3): The IP protocol helps route data packets from my device toward the destination through the internet.

* Data Link Layer (Layer 2): Wi-Fi or Ethernet carries data across the local network between my device and the router.

* Physical Layer (Layer 1): Radio signals transmit data between my phone and the Wi-Fi router.

4. Given this scenario: You are unable to access a website, but you can successfully ping its IP address. Identify which OSI layer(s) might be causing the problem and explain your reasoning.

Ans:-

Possible OSI Layers Causing the Problem:

* Application Layer (Layer 7): The website or web server may be down, or the web service may not be working correctly.

* Presentation Layer (Layer 6): There may be an SSL/TLS certificate or encryption problem preventing a secure HTTPS connection.

* Session Layer (Layer 5): A problem establishing or maintaining the connection session may prevent access to the website.

* Transport Layer (Layer 4): Ports 80 (HTTP) or 443 (HTTPS) may be blocked by a firewall, or the web server may not be listening on the required port.

Reasoning: Since the website's IP address responds to ping, IP connectivity is working at least partially. The problem may be related to the web service, TCP ports, firewall rules, or HTTPS configuration.

Conclusion: Check the Application Layer (Layer 7) and Transport Layer (Layer 4) first, then investigate DNS or TLS if necessary.